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Stokes and upconverted luminescence in
 $\text{Er}^{3+}/\text{Yb}^{3+}$ -doped $\text{Y}_3\text{Ga}_5\text{O}_{12}$ nano-garnets

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The green, red, near-infrared and near-infrared-to-visible upconverted luminescence properties of $\text{Er}^{3+}/\text{Yb}^{3+}$ codoped $\text{Y}_3\text{Ga}_5\text{O}_{12}$ nanocrystalline powders have been studied using laser spectroscopy. A diffuse reflectance and luminescence spectra confirm that Er^{3+} and Yb^{3+} ions occupy the Y^{3+} sites of the single-phase cubic nano-garnet. Very bright green and red luminescence of the Er^{3+} ions are detected by the naked eyes, even for a laser power as low as 15 mW, when the Yb^{3+} ions are excited at 970 nm. The red upconverted emission is more intense than that under direct excitation of the Er^{3+} ions. The power dependence and the dynamics of the near-infrared-to-green and near-infrared-to-red upconverted emissions show the existence of different two-photon energy transfer upconversion processes. The results here presented indicate that $\text{Er}^{3+}/\text{Yb}^{3+}$ codoped $\text{Y}_3\text{Ga}_5\text{O}_{12}$ can be a good candidate as an optical nanoheater and nanothermometer in biomedicine applications in the first biological window.

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Introduction

Trivalent rare-earth (RE^{3+}) activated upconversion (UC) nano-crystals are attractive due to their interesting physicochemical features, such as large anti-Stokes shifts, low autofluorescence background, low toxicity and high penetration depth. These features make them future alternatives to conventional organic dyes and quantum dots for various photonic and biomedical applications, for instance light converters¹ white light generation,² optical thermometry,^{3,4} bio-imaging,⁵ drug delivery,⁶ and photothermal therapy⁷ *etc.* A complete control of the photophysical (structure *vs.* luminescence) properties of these systems is highly necessary for their design and development. The RE^{3+} -doped UC nanoparticles still suffer from a low quantum yield, which is currently the major drawback for their commercialization. On the other hand, obtaining nanocrystalline RE^{3+} -doped crystals doped with good luminescence properties is a challenging task. In this direction, recent advances in the synthesis of nano-crystals by means of novel approaches allow nowadays both the control of their structure-morphology and their luminescence properties.^{8,9}

RE^{3+} -doped UC nanoparticles are able to convert low energy (higher wavelength) radiation into high energy (shorter wavelength) emission, due to their unique electronic structures that enable to emit photons from high energy levels through successive photon absorption and energy transfer (ET) processes from their low-lying energy levels.¹⁰ Nanocrystalline garnets doped with RE^{3+} ions allow obtaining tunable properties based on crystalline size and shape with well-known synthesis techniques.^{8,9} Moreover, the nano-garnets are good hosts for RE^{3+} ions due to their low phonon energy, good infrared transparency, high refractive index (higher than 2) and good chemical stability.¹¹ Many studies on the RE^{3+} -doped nano-garnets demonstrate that they are excellent hosts for the lanthanides. For instance, Nd^{3+} -doped $\text{Ca}_3\text{Ga}_2\text{Ge}_3\text{O}_{12}$ garnet has been presented as a new optical pressure sensor because it has two intense and isolated lines in the near-infrared (NIR) interval from 926 to 943 nm, which are more sensitive to pressure than the R1 line of Cr^{3+} in ruby, the most widely used pressure sensor in high-pressure experiments.¹² On the other hand, the Dy^{3+} -doped $\text{Lu}_3\text{Ga}_5\text{O}_{12}$ (LuGG) has been proposed as a suitable candidate for diodes due to its ability to address white light by the sum of blue and yellow emissions by monochromatic laser excitation.¹³

Concretely, the Re^{3+} -doped $\text{Y}_3\text{Ga}_5\text{O}_{12}$ (YGG) nano-garnet presents significant luminescent properties. On the one hand, the Er^{3+} -doped YGG nano-garnet exhibits a relative sensitivity much higher than other samples also proposed as nano-sensors to measure the temperature in biological systems.⁴ Lozano-Gorrín *et al.*¹⁴ studied the Nd^{3+} single-doped and the $\text{Er}^{3+}/\text{Yb}^{3+}$ co-doped YGG nano-garnets for their potential use

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as simultaneous optical nanoheaters and nanothermometers in biomedicine applications in the first biological window. On the other hand, Hua *et al.*¹⁵ reported luminescent properties of transparent Dy^{3+} -doped YGG and Cr^{3+} -doped YGG ceramics for white light emitting devices and Pathak *et al.*¹⁶ reported a narrow band ultraviolet B emission (312 nm) from Gd^{3+} -doped YGG nano-garnets for phototherapy applications.

Despite being a widely studied compound, to the best of our knowledge, there are no reports on the Stokes and UC luminescence properties of $\text{Er}^{3+}/\text{Yb}^{3+}$ co-doped YGG nano-garnets. The present investigation is aimed to study the effect of Yb^{3+} ions on Stokes and anti-Stokes luminescence properties of Er^{3+} ions in YGG nano-garnets prepared by a low-cost citrate sol-gel¹¹ method.

Experimental details

Three nanocrystalline garnets with stoichiometry $(\text{Y}_{1-x-y}\text{Er}_x\text{Yb}_y)_3\text{Ga}_5\text{O}_{12}$, where $x = 0.001$ and 0.01 for $y = 0$ (labeled by YGG01Er and YGG1Er), and $x = 0.01$ for $y = 0.1$, (denoted as YGG1Er10Yb), were prepared by a low-cost citrate sol-gel method.¹¹ First, high-purity $\text{Ga}(\text{NO}_3)_3 \cdot 9\text{H}_2\text{O}$, $\text{Y}(\text{NO}_3)_3 \cdot 4\text{H}_2\text{O}$, $\text{Er}(\text{NO}_3)_3 \cdot 5\text{H}_2\text{O}$ and $\text{Yb}(\text{NO}_3)_3 \cdot 5\text{H}_2\text{O}$ materials were disintegrated in 25 ml of 1 M HNO_3 by stirring at 353 K for 3 h. Second, citric acid was incorporated to the solution and this was stirred for 2 h. Third, samples were dried at 363 K for 36 h to create the gel phase. The latter was fired at 773 K for 4 h to eliminate the extra nitrates and organic mixtures. Finally, the powder compound was annealed at 1173 K for 16 h for the creation of nanocrystalline particles. The structure of the resultant sample was studied by X-ray powder diffraction (XRD) by using the PANalytical X'Pert Pro equipment with $\text{CuK}\alpha 1$ radiation ($\lambda = 1.5406 \text{ \AA}$) (see Fig. 1).

The visible and NIR diffuse reflectance spectrum in the range from 470 to 1660 nm was collected with the Varian Cary 6000i spectrophotometer. The luminescence of the Er^{3+} and $\text{Er}^{3+}/\text{Yb}^{3+}$ -doped YGG nano-garnets were obtained by exciting at 488 nm with an Ar^+ laser (Spectra Physics 2060-10 Beamlock) for the visible range and by exciting at 650 nm with a diode laser for the NIR range. Visible emissions were concentrated in a fiber coupled 0.303 m single-grating Andor Shamrock SR-303i-B spectrograph and then they were collected by a cooled CCD detector. NIR emissions were focalized onto an Andor Shamrock SR-500i-B2 monochromator with a resolution of 0.1 nm and then an InGaAs (Andor iDUS) photodiode array was used to detect the signal. Moreover, the green and red UC luminescence were measured under a cw 970 nm laser excitation by using a Ti:sapphire tunable laser (3900S Spectra Physics) pumped by a 532 nm diode laser (Spectra Physics Millennia Prime Laser), with the same setup used for the visible range. Stokes and upconverted luminescence decay curves were collected with a 10 ns pulsed optical parametric oscillator OPO (EKSPALA/NT342/3/UVE) by exciting the Er^{3+} and Yb^{3+} ions, respectively. The signal was observed through a digital oscilloscope (LeCroy WS424) coupled to the detector.

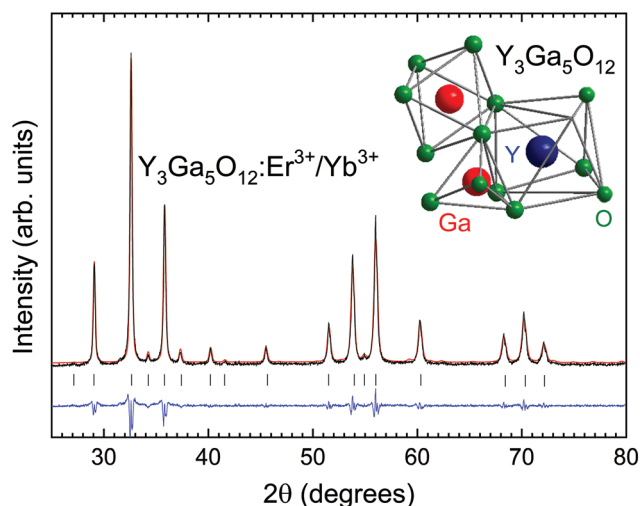


Fig. 1 X-ray diffraction pattern of the $\text{Y}_3\text{Ga}_5\text{O}_{12}$ nano-garnet doped with Er^{3+} and Yb^{3+} ions. Le Bail fitting (in red), the difference between the observed and the calculated pattern (in blue) and the reflections for this material (vertical lines) are also included. Inset shows interconnected octahedral, tetrahedral and dodecahedral sites of garnet structure.

The spectral response of the equipment was eliminated from all luminescence curves.

Results and discussion

Nano-crystalline structure

The XRD diffraction pattern of the YGG nano-garnet co-doped with Er^{3+} and Yb^{3+} optically active ions, presented in Fig. 1, shows a series of well-defined Bragg reflections as a fingerprint of a crystalline structure. These reflections have been indexed to the YGG garnet structure with a body-centered cubic (bcc) structure (space group $Ia\bar{3}d$, No. 230, $Z = 8$), and refined by Le Bail method ($\chi^2 = 4.95$, $R_p = 7.98$; $R_{wp} = 10.5$; $R_{exp} = 4.66$) using the FULLPROF program.¹⁷ The cubic lattice parameter is of $12.274(1) \text{ \AA}$ obtained by Le Bail fitting and the nano-crystalline size is of $33(8) \text{ nm}$ according to the Scherrer equation, $\beta(2\theta) = K \lambda / L \cos(\theta)$, where β is the line broadening at half of the maximum intensity (FWHM), K is a constant ~ 1 and L is the crystalline size. The Ga atoms occupy two different Wyckoff positions (WP), 16a and 24d, with octahedral (C_{3i}) and tetrahedral (S_4) symmetry, respectively. The Y atoms are in the WP 24c with dodecahedral (D_2) symmetry and O atoms are in the WP 96 h characterized by the three free structural parameters (x, y, z). Such a structure is a network of interconnected dodecahedral (YO_8), octahedral (GaO_6) and tetrahedral (GaO_4) units with shared oxygen atoms (Fig. 1). The Yb^{3+} and Er^{3+} ions will predominantly substitute the Y^{3+} ions and therefore possess D_2 local point symmetry.¹⁸

Diffuse reflectance and luminescence properties

The optical properties of the RE^{3+} ions depend on the local structure where they are located in the host network.¹⁹ The

crystal-field, formed by the RE^{3+} ion and its oxygen ligands, causes the fine structure splitting of the free-ion $2S+1L_J$ multiplets and the intra-configurational 4f–4f electric-dipole transitions probabilities in the optical (UV-Vis-NIR) range. As commented above, the Yb^{3+} and Er^{3+} ions substitute the Y^{3+} ions with D_2 symmetry in the nano-garnet. Such a symmetry leads to a splitting of the degenerated levels that gives rise to the $(2J + 1)/2$ Stark, or crystal-field, levels.¹⁹

The diffuse reflectance spectra of the YGG1Er and YGG1Er10Yb nano-garnets in the visible-NIR range are given in Fig. 2. The spectra show sharp structured bands confirming the incorporation of the RE^{3+} in the nanocrystalline garnet structure. These bands are associated to intra-configurational $4f^N-4f^N$ electronic transitions between the $^4\text{I}_{15/2}$ ground state and the different excited levels for the Er^{3+} ions, and between the $^2\text{F}_{5/2}$ ground state and the $^2\text{F}_{7/2}$ unique excited multiplet of the Yb^{3+} ions. The nature of all these transitions is electric dipole, excluding that to the $^4\text{I}_{13/2}$ first excited multiplet of the Er^{3+} ions, which also exhibits an important magnetic dipole contribution.²⁰ A simplified energy level scheme of the Er^{3+} and Yb^{3+} ions in the YGG nano-garnet are represented in Fig. 3a.²¹

When the Er^{3+} ions in the YGG1Er nano-garnet are resonantly excited by a blue laser radiation (488 nm) to the $^4\text{F}_{7/2}$ level, the $^2\text{H}_{11/2}$ and $^4\text{S}_{3/2}$ thermalized levels are populated by non-radiative multiphonon relaxation processes and a bright green emission in the 520–560 nm range is measured. In addition, weak emissions in the red and the far-red ranges are also observed around 640–680 nm, corresponding to the $^4\text{F}_{9/2} \rightarrow ^4\text{I}_{15/2}$ transition, and at 840–860 nm, assigned to the $^4\text{S}_{3/2} \rightarrow ^4\text{I}_{13/2}$ transition (see Fig. 4a). Furthermore, under 650 nm red laser radiation, NIR emissions were also measured at around 1020 nm, associated with the $^4\text{I}_{11/2} \rightarrow ^4\text{I}_{15/2}$ transition, and in the 1460–1660 nm range, corresponding to the $^4\text{I}_{13/2} \rightarrow ^4\text{I}_{15/2}$ transition (see Fig. 4b). The $^4\text{F}_{9/2}$, $^4\text{I}_{11/2}$, and $^4\text{I}_{13/2}$ multiplets are populated through different resonant and quasi-resonant

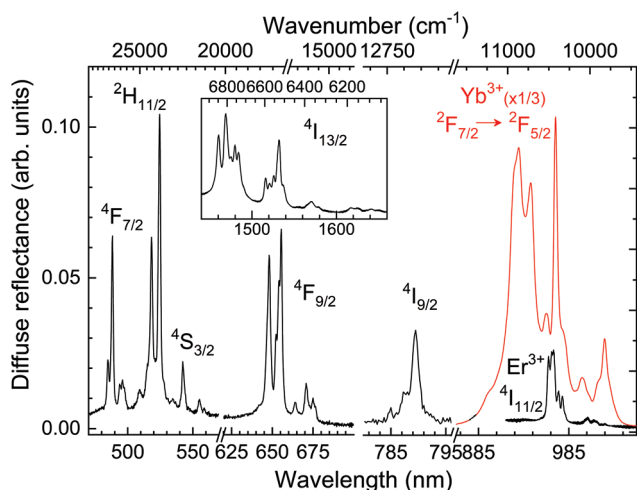


Fig. 2 Diffuse reflectance spectra of the YGG1Er and YGG1Er10Yb nano-garnets in the visible-NIR range at room temperature.

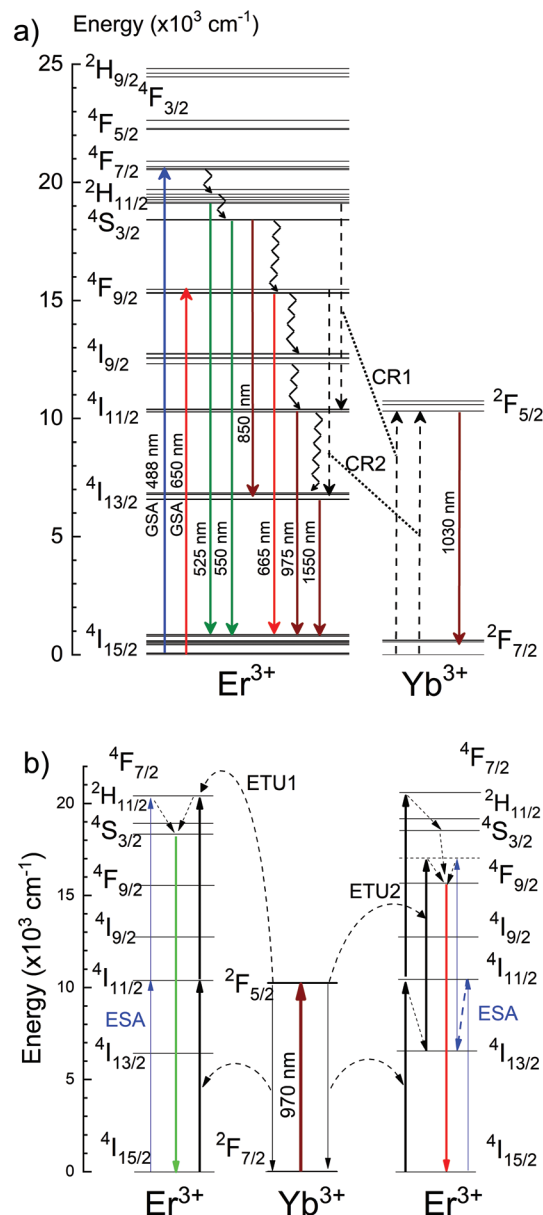


Fig. 3 (a) Schematic representation of energy levels involved in the emission of YGG1Er10Yb under an excitation of 488 nm and 650 nm. Ground state absorption (GSA), photoluminescence (solid color lines), multiphonon (wavy lines) and energy transfer channels (dotted lines) are included. (b) Schematic representation of the energy transfer UC processes in the YGG1Er10Yb nano-garnet under 970 nm NIR cw laser excitation.

non-radiative de-excitation mechanisms, *i.e.* multiphonon and/or ET cross-relaxation (CR) channels between the Er^{3+} ions. For the YGG1Er10Yb nano-garnet, the ET mechanisms must also involve non-radiative interactions between the Er^{3+} donor ions and the Yb^{3+} acceptor ions in order to explain the $^2\text{F}_{5/2} \rightarrow ^2\text{F}_{7/2}$ emission band of the Yb^{3+} ion, centered at around 1030 nm. It is worthy remarking the enhancement of the emission at 1030 nm due to this latter transition. In order to observe this increase of intensity, the NIR emission spectra

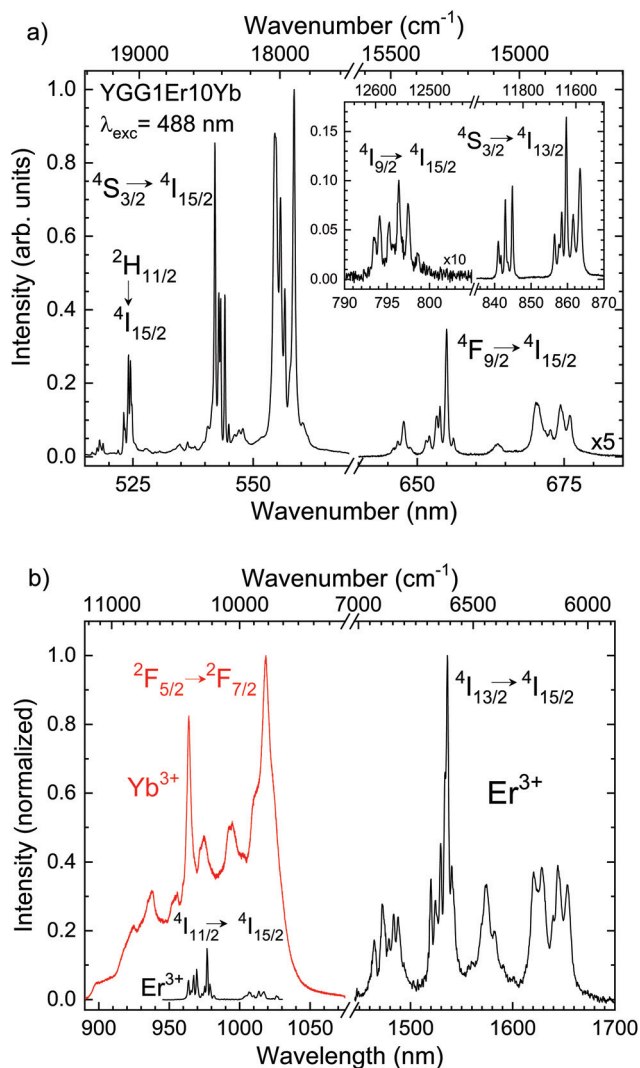


Fig. 4 (a) Visible luminescence spectra exciting with a laser light at 488 nm and (b) near-infrared luminescence spectra exciting under 650 nm laser radiation of the YGG1Er10Yb nano-garnet. The emission bands of the Yb³⁺ ions centred around 980 nm and of the Er³⁺ ion centred around 1550 nm have been normalized for comparison. The emission of the 1% Er³⁺ single-doped nano-garnet associated with the $^4F_{9/2} \rightarrow ^4I_{15/2}$ transition has been also included for comparison.

of YGG1Er and YGG1Er10Yb have been compared in Fig. 4b after normalizing with respect to the emission of 1535 nm in both cases.

Under direct excitation of the Er³⁺ ions at 488 nm, the green/red intensity emission ratio of the green emission to the red one is found to be around 7:1, showing the high efficiency of the $^2H_{11/2}$ and $^4S_{3/2}$ thermalized levels, with high spontaneous emission probabilities compared to the multiphonon relaxation probability, that feeds the $^4F_{9/2}$ next lower level. Here, it is important to mention that the sharp peak profiles of the emission bands further confirm that the Er³⁺ and Yb³⁺ ions are incorporated in the distorted dodecahedra with D_2 symmetry of the YGG nano-garnet crystal.

The existence of ET channels in the YGG nano-garnets doped with Er³⁺ and co-doped with Yb³⁺ ions can be easily checked analyzing the luminescence decay curves, and hence the lifetimes, of the green and red emitting levels. The luminescence decay curves of the ($^2H_{11/2}$, $^4S_{3/2}$) and $^4F_{9/2}$ multiplets of the Er³⁺ ions in the YGG nano-garnets doped with 0.1% and 1% of Er³⁺ ions, as well as codoped with 10% of Yb³⁺ ions, under a 525 and 655 nm pulsed laser excitations, respectively, are presented in the Fig. 5. When the YGG nano-garnet is only doped with a low concentration of 0.1 mol% of Er³⁺ ions, a homogeneous distribution of optically active ions in the garnet cubic structure is expected. A large enough Er³⁺–Er³⁺ distance avoids the existence of effective ET mechanisms affecting the de-excitation of Er³⁺ ions in the ($^2H_{11/2}$, $^4S_{3/2}$) thermalized levels. This fact is confirmed by the single exponential nature of the green and red luminescence decay curves, indicating that they are only affected by the radiative and non-radiative multiphonon relaxation probabilities. However, the decay curve of these levels shows non-exponential behaviors when increasing the Er³⁺ concentration to 1 mol% and codoping with 10 mol% of Yb³⁺ ions, indicating that the ET mechanisms produce the relaxation of the Er³⁺ ions and generates the Yb³⁺ NIR emission.

The intrinsic lifetime τ_0 of the ($^2H_{11/2}$, $^4S_{3/2}$) thermalized levels, calculated from the exponential decay curve of the YGG nano-garnet with lowest doping concentration, is 130 μ s. At a higher Er³⁺ concentration of 1 mol%, the luminescence decay has a non-exponential character and it is accompanied by a reduction of the lifetime to 100 μ s, which is associated to ET mechanisms between Er³⁺ ions and/or to different quenching traps (impurities and/or defective sites)²² and has been calculated using the effective lifetime equation, *i.e.* $\tau_{\text{eff}} = \int t \cdot I(t) dt / \int I(t) dt$. The thermalized levels are depopu-

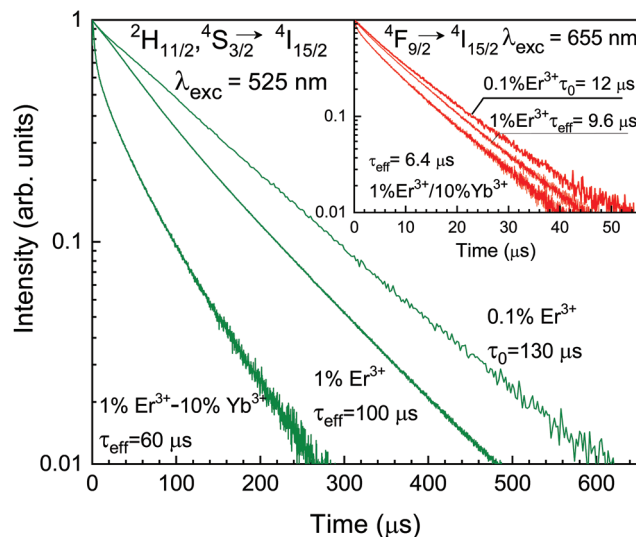


Fig. 5 The luminescence decay curve of the $^2H_{11/2}$ and $^4S_{3/2}$ thermalized levels of the Er³⁺ ions obtained exciting at 525 nm pulsed laser excitation. Inset shows the luminescence decay curve of the $^4F_{9/2}$ red emitting level of the Er³⁺ ions under 655 nm laser excitation.

lated by these resonant ($^4S_{3/2}, ^4I_{15/2} \rightarrow ^4I_{13/2}, ^4I_{9/2}$) and ($^4S_{3/2}, ^4I_{15/2} \rightarrow ^4I_{9/2}, ^4I_{13/2}$) cross-relaxation channels, labeled as I and II in Fig. 3a. The first channel involves one Er^{3+} ion excited at the $^4S_{3/2}$ emitting level and the other one at the $^4I_{15/2}$ ground state. The excited Er^{3+} ion emits to the $^4I_{13/2}$ first excited multiplet and such a transition energy is transferred non-radiatively to the other Er^{3+} ion that is excited from the $^4I_{15/2}$ ground state to the $^4I_{9/2}$ level. The second channel takes place between different levels, but the point is that both channels finally feed the $^4I_{11/2}$ level, after a fast non-radiative relaxation from the $^4I_{9/2}$ level, and the $^4I_{13/2}$ lowest emitting level.

The Er^{3+} - Er^{3+} ET rate of the green emitting levels can be estimated for the higher Er^{3+} concentration of 1 mol% using the equation $\tau_{eff}^{-1} = \tau_0^{-1} + W_{ET}$, giving a value of $W_{ET} = 2308 \text{ s}^{-1}$. This rate is lower, but comparable, than the intrinsic (radiative plus multiphonon) probability of ($\tau_0^{-1} = 7692 \text{ s}^{-1}$) obtained for the ($^2H_{11/2}, ^4S_{3/2}$) thermalized levels.

This estimation allows concluding that the ET probability starts to be significant in the quenching of the green luminescence from concentrations of 1 mol% of Er^{3+} ions.

NIR-to-visible energy upconversion

In this section we analyse the NIR-to-visible upconverted luminescence for the YGG1Er10Yb nano-crystal. Before that, it is convenient to explain the two possible mechanisms responsible for the population of the excited emitting level. On the one hand, the excited state absorption (ESA) where only one optically active ion is promoted to the upper levels by the resonant absorption of two or more laser photons. On the other hand, the ET upconversion (ETU) where two or more nearby interacting optically active ions are involved. Both mechanisms could coexist but one of them usually dominates the upconversion process.¹¹

The UC luminescence of the YGG1Er10Yb nano-crystal can be detected by the naked eyes excited when a laser light at 970 nm, even for a laser power as low as 15 mW. This excitation is resonant with the $^4I_{15/2} \rightarrow ^4I_{11/2}$ transition of Er^{3+} as well as with the $^2F_{5/2} \rightarrow ^2F_{7/2}$ transition of Yb^{3+} (Fig. 2), giving rise to multiple possibilities for the UC processes. The UC spectra, shown in Fig. 6, consist of the $^2H_{11/2}, ^4S_{3/2} \rightarrow ^4I_{15/2}$ green and $^4F_{9/2} \rightarrow ^4I_{15/2}$ red emissions of the Er^{3+} ions. The difference between the oscillator strengths of the single doped (YGG1Er) and the co-doped (YGG1Er10Yb) samples is significant, the latter being 6 times larger. This difference would indicate that the Yb^{3+} ions are more effectively pumped and, consequently, a large contribution of the ETU mechanism associated to Yb^{3+} to Er^{3+} ET processes over ESA for the UC emission is expected. However, the ESA and ETU mechanisms among Er^{3+} ions cannot be disregarded.

In order to analyze in more detail, the ETU mechanisms involved in the Yb^{3+} to Er^{3+} transfer processes, the intensity dependences of the green and red UC emissions with the laser pump power after the 970 nm excitation have been obtained. It is known that the intensity of the upconverted luminescence (I_{UP}) is proportional to n^{th} power of the pumped excitation intensity, being n the number of photons involved in UC

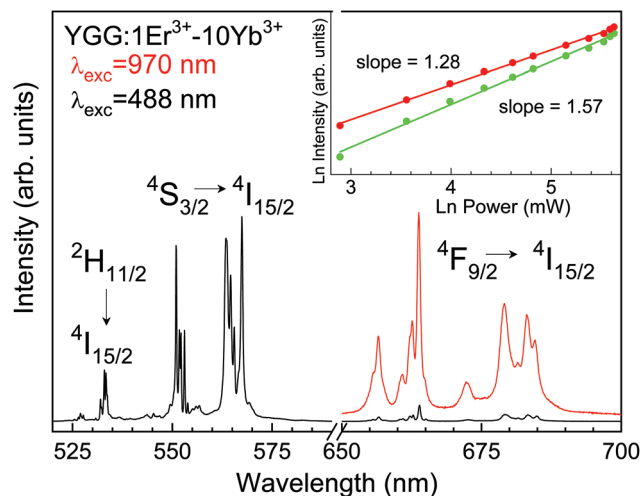


Fig. 6 Comparison between Stokes and anti-Stokes luminescence exciting at 488 and 970 nm respectively. Inset shows the variation of the green (550 nm) and the red (650 nm) UC emission intensities as a function of the pump power.

mechanism (see inset of Fig. 6).²³ The values of the power dependence curves are $n = 1.28$ for the red UC emission and $n = 1.57$ for the green one, indicating that the ETU mechanism is the main responsible for these two UC emissions, needing an energy equivalent to two laser photons.

Differences between the emission spectra for direct excitation and UC were observed (Fig. 6). In this sense, it is worth noting that the ratio of the green-to-red emission is about 0.8:1 in the NIR-to-visible UC spectra exciting at 970 nm, reflecting that the red UC emission is around 9 times larger than that obtained under direct excitation of the Er^{3+} ions at 488 nm and it is slightly more intense than the green UC luminescence (see Fig. 6). According to the later, it is obvious that the multiphonon relaxation from the thermalized levels is insufficient to populate the $^4F_{9/2}$ level. This significant enhancement in the red emission in the UC spectra has to be ascribed to a different and highly efficient non-resonant ETU channel that feeds the red emitting level, independently of the one that populates the ($^2H_{11/2}, ^4S_{3/2}$) thermalized levels. In order to understand this, the dynamics of the green and red luminescence of Er^{3+} ions were measured under 10 ns pulse excitation at 970 nm. The two different decay times are clearly observed in the temporal evolutions for the green and red UC emissions. The rise time of the red UC luminescence is slightly faster than that for the green UC emission. The temporal evolution of both UC emissions not only confirms that the most probable mechanism for the UC processes is ETU but also it is an evidence for the existence of a second efficient ETU channel that feeds the $^4F_{9/2}$ red emitting level.

The ET UC mechanisms are illustrated in Fig. 3b. The initial 970 nm laser photon will populate the $^2F_{5/2}$ level of the Yb^{3+} ions and the $^4I_{11/2}$ level of Er^{3+} ions. The Yb^{3+} ions will transfer its energy to a nearby Er^{3+} ion in the ground

state to be promoted to the $^4I_{11/2}$ level or to an excited nearby Er^{3+} ion. In the first case, the transfer of energy from a second excited nearby Yb^{3+} ion will promote an excited Er^{3+} ion to its $^4F_{7/2}$ level (ETU1), from which they are non-radiatively de-excited to $^2H_{11/2}$, $^4S_{3/2}$ thermalized level and to $^4F_{9/2}$ red emitting level *via* multiphonon relaxation, followed by a radiative decay giving rise to the strong green emission in the 525–590 nm region due to the $^2H_{11/2}$, $^4S_{3/2} \rightarrow ^4I_{15/2}$ transition and in the red range around 665 nm associated to the $^4F_{9/2} \rightarrow ^4I_{15/2}$ transition.

In the second upconversion process in both previous cases, the population of Er^{3+} ions in the $^4I_{11/2}$ level is followed by a non-radiative multiphonon relaxation to the $^4I_{13/2}$ one, from which a second resonant $Yb^{3+} \rightarrow Er^{3+}$ transfer will populate the $^4F_{9/2}$ level giving red emission through $^4F_{9/2} \rightarrow ^4I_{15/2}$ transition (ETU2). The $^4I_{13/2} \rightarrow ^4F_{9/2}$ transition has a lower energy (8700 cm^{-1}) than this associated to the $^2F_{7/2} \rightarrow ^2F_{5/2}$ transition (10309 cm^{-1}) with mismatch of around 1609 cm^{-1} . This suggests that the ET between Yb^{3+} and Er^{3+} ions is not resonant but because of host phonon assisted.

Finally, it is important to mention that there is a looping process since part of the energy of the Er^{3+} ions involved in the relaxation from thermalized levels will feed the excited $^2F_{5/2}$ level of the Yb^{3+} ions *via* an $Er^{3+} \rightarrow Yb^{3+}$ cross relaxation process, due to the high concentration of Yb^{3+} ions in the YGG nano-garnet.

Conclusions

The structure of peaks observed in the absorption and emission spectra of the YGG1Er and YGG1Er10Yb nano-garnets confirms that the Er^{3+} and Yb^{3+} ions are incorporated to dodecahedra with D_2 symmetry. The non-exponential behavior of the luminescence decay curve of the $^2H_{11/2}$, $^4S_{3/2}$ thermalized levels and of the $^4F_{9/2}$ red emitting level of Er^{3+} ions shows that there is ET between Er^{3+} ions for the YGG1Er sample and also between Er^{3+} and Yb^{3+} ions in the co-doped nano-crystal. The UC luminescence of Er^{3+} ions has been measured and it is worth noting that the red UC emission has been enhanced by around 9 times due to two channels that feed the red emitting level. One of them comes *via* multiphonon relaxation from $^2H_{11/2}$, $^4S_{3/2}$ levels and the other corresponds to a very efficient non-resonant ETU mechanism that populates the $^4F_{9/2}$ level. The green and red UC emission of the Er^{3+}/Yb^{3+} co-doped crystalline powder is clearly visible to the naked eye for laser power as low as 15 mW, which makes it strongly interesting for its use as an UC phosphor, UC lasers and optical labels for biological studies in the green and red region under NIR excitation.

Author contributions

Experiments were carried out by VM and VV. Discussion and writing have been performed by VM, VV, VL and URRM.

Conflicts of interest

There are no conflicts to declare.

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